4.1 $\tilde{y}_i = \tilde{b}_0 + \tilde{b}_1 x_i$, residual becomes $\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1 \tilde{x}_i = 0$

$$\frac{1}{n}\sum_{i=1}^{n} y_i - \bar{y}$$

mean squared error: $\frac{1}{n}\sum_{i=1}^{n}(\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1\tilde{x}_i)^2$

so add $+a(b_1)^2$

$$\min_{\tilde{b}_1}\left[\frac{1}{n}\sum_{i=1}^{n}(\tilde{y}_i - b_0 - \tilde{b}_1\tilde{x}_i)^2 + a(\tilde{b}_1)^2\right]$$

4.2

$$L(\tilde{b}_0, \tilde{b}_1) = \frac{1}{n}\sum_{i=1}^{n}(\tilde{y}_i - b_0 - \tilde{b}_1\tilde{x}_i)^2 + a(\tilde{b}_1)^2$$

$$\frac{dL}{d\tilde{b}_0} = \frac{1}{n}\sum_{i=1}^{n} 2(\tilde{y}_i - b_0 - \tilde{b}_1\tilde{x}_i)(-1)$$

$$= -\frac{2}{n}\sum_{i=1}^{n}(\tilde{y}_i - b_0 - \tilde{b}x_i) = 0$$

$$\frac{dL}{d\tilde{b}_1} = \frac{1}{n}\sum_{i=1}^{n} 2(\tilde{y}_i - b_0 - \tilde{b}_1\tilde{x}_i)(-\tilde{x}_i) + 2ab_1$$

$$= -\frac{2}{n}\sum_{i=1}^{n}\tilde{x}_i(\tilde{y}_i - b_0 - \tilde{b}_1\tilde{x}_i) + 2ab_1 = 0$$

Assumes $\sum \tilde{x}_i = 0$ & $\sum \tilde{y}_i = 0$ from mean-centering from normalizing

$$\sum_i(\tilde{y}_i - \tilde{b}_0 - \tilde{b}_1\tilde{x}_i) = 0$$

$$\sum \tilde{y}_i - n\tilde{b}_0 - \tilde{b}_1\sum \tilde{x}_i = 0$$

$$0 - n\tilde{b}_0 - 0 = 0 \therefore \tilde{b}_0 = 0$$

$$-\sum \tilde{x}_i\tilde{y}_i + \tilde{b}_1\sum \tilde{x}_i^2 + na\tilde{b}_1 = 0$$

$$\tilde{b}_1\left(\sum \tilde{x}_i^2 + na\right) = \sum \tilde{x}_i\tilde{y}_i$$

$$\tilde{b}_1 = \frac{\sum \tilde{x}_i\tilde{y}_i}{\sum_i \tilde{x}_i^2 + n\alpha}$$

4.3 The slope decreases b/c in $\tilde{b}_1$, α is in denominator.

4.4

$L(b_0, b_1) = \frac{1}{n}\sum_{i=1}^{n}(\tilde{y}_i - b_0 - b_1\tilde{x}_i)^2 + a|b_1|$ & $b_0 = 0$ b/c centering

$L(b_1) = \frac{1}{n}\sum_{i=1}^{n}(\tilde{y}_i - b_1\tilde{x}_i)^2 + a|b_1|$

Assume $b_1 > 0$,

$$\frac{dL}{db_1} = -\frac{2}{n}\sum_{i=1}^{n}\tilde{x}_i(\tilde{y}_i - b_1\tilde{x}_i) + a = 0 \quad \therefore b_1 = \frac{\sum \tilde{x}_i\tilde{y}_i - \frac{na}{2}}{\sum \tilde{x}_i^2}$$

We cant do $b_1 = \frac{\sum \tilde{x}_i\tilde{y}_i + \frac{na}{2}}{\sum \tilde{x}_i^2}$ b/c $b_1 < 0$

$b_1 = 0$ optimal when $\left|\sum \tilde{x}_i\tilde{y}_i\right| \le \frac{na}{2}$ b/c we can't

do derivative there so do hella small so penalty can just make it 0